IMPURITIES ASCEND ALONG A FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an etale chart construction of the fibre product, a lift of the specialisation along the flat induced morphism, and a closure comparison using base change of the relative assassin

Editorial note. The label lemma-flat-ascent-impurity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Impurities above a point, and the base change of the relative assassin of a module
- The existence of an etale chart of a fibre product through a prescribed pair of points, and the resulting flatness of the induced morphism of schemes
- Lifting of specialisations along a flat morphism of schemes, and the compatibility of the closed subspace cut out by a point with the base change

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Prerequisite pointer invoked at proof line(s) 387; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05FW: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 260–260. Prerequisite pointer invoked at proof line(s) 367; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CV2: sources/stacks-chow/tags.txt, source lines 16122–16122. Prerequisite pointer invoked at proof line(s) 393; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03MO: sources/stacks-chow/tags.txt, source lines 4756–4756. Prerequisite pointer invoked at proof line(s) 383; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Prerequisite pointer invoked at proof line(s) 381; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-flat.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 05FW. Let $(Y_1, y_1) \to (Y, y)$ be a morphism of pointed algebraic spaces over S. Assume $Y_1 \to Y$ is flat at y_1. If $(T \to Y, t' \rightsquigarrow t, \xi)$ is an impurity of $\mathcal{F}$ above y, then there exists an impurity $(T_1 \to Y_1, t_1' \rightsquigarrow t_1, \xi_1)$ of the pullback $\mathcal{F}_1$ of $\mathcal{F}$ to $X_1 = Y_1 \times_Y X$ over y_1 such that T_1 is étale over $Y_1 \times_Y T$.*

Proof. Choose an étale morphism $T_1 \to Y_1 \times_Y T$ where T_1 is a scheme and let $t_1 \in T_1$ be a point mapping to y_1 and t. It is possible to find a pair (T_1, t_1) like this by Properties of Spaces, Lemma 03H4. The morphism of schemes $T_1 \to T$ is flat at t_1 (use Morphisms of Spaces, Lemma 03MO and the definition of flat morphisms of algebraic spaces) there exists a specialization $t_1' \rightsquigarrow t_1$ lying over $t' \rightsquigarrow t$, see Morphisms, Lemma 03HV. Choose a point $\xi_1 \in |X_{T_1}|$ mapping to t_1' and ξ with $\xi_1 \in \mathrm{Ass}_{X_{T_1}/T_1}(\mathcal{F}_{T_1})$. point of $\mathrm{Spec}(\kappa(t_1') \otimes_{\kappa(t')} \kappa(\xi))$. This is possible by Divisors on Spaces, Lemma 0CV2. As the closure Z_1 of $\{\xi_1\}$ in $|X_{T_1}|$ maps into the closure of $\{\xi\}$ in $|X_T|$ we conclude that the image of Z_1 in $|T_1|$ cannot contain t_1. Hence $(T_1 \to Y_1, t_1' \rightsquigarrow t_1, \xi_1)$ is an impurity of $\mathcal{F}_1$ above Y_1. □